

AUTHENTICATION REQUIREMENTS

- Disclosure
- Traffic analysis
- Masquerade
- Content modification
- Sequence modification
- Timing modification
- Source repudiation
- Destination repudiation

Disclosure

- Release of message contents to any person or process not possessing the appropriate cryptographic key.

Traffic analysis

- Discovery of the pattern of traffic between parties.
- In a connection oriented application, the frequency and duration of connections could be determined.
- In either a connection-oriented or connectionless environment, the number and length of messages between parties could be determined.

Masquerade:

- Insertion of messages into the network from a fraudulent source.
- includes the creation of messages by an opponent that are purported to come from an authorized entity.
- Also included are fraudulent acknowledgments of message receipt or non receipt by someone other than the message recipient.

Content modification

- Changes to the contents of a message, including insertion, deletion, transposition, and modification.

Sequence modification

- Any modification to a sequence of messages between parties, including insertion, deletion, and reordering.

Timing modification

- Delay or replay of messages.
- In a connection-oriented application, an entire session or sequence of messages could be a replay of some previous valid session, or individual messages in the sequence could be delayed or replayed.
- In a connectionless application, an individual message (e.g:- datagram) could be delayed or replayed.

- **Source repudiation**

- Denial of transmission of message by source.

- **Destination repudiation:**

- Denial of receipt of message by destination.
- Message authentication is a procedure to verify that received messages come from the alleged source and have not been altered.
- Message authentication may also verify sequencing and timeliness.

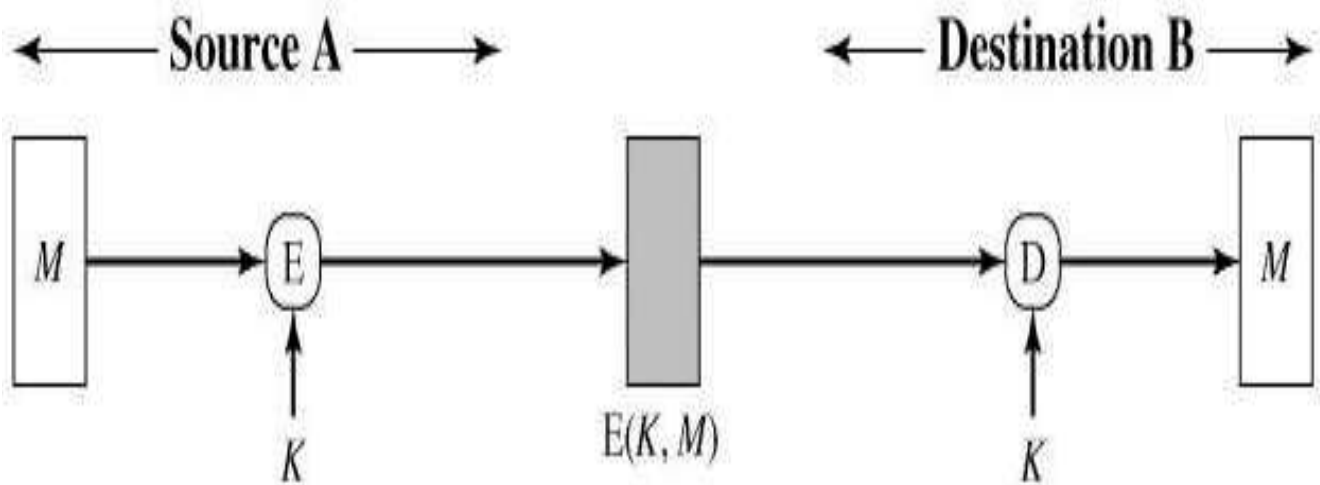
AUTHENTICATION FUNCTIONS

- The types of functions that may be used to produce an authenticator may be grouped into three classes
 - Message encryption
 - Message authentication code (MAC)
 - Hash function:

(1) Message encryption - The cipher text of the entire message serves as its authenticator.

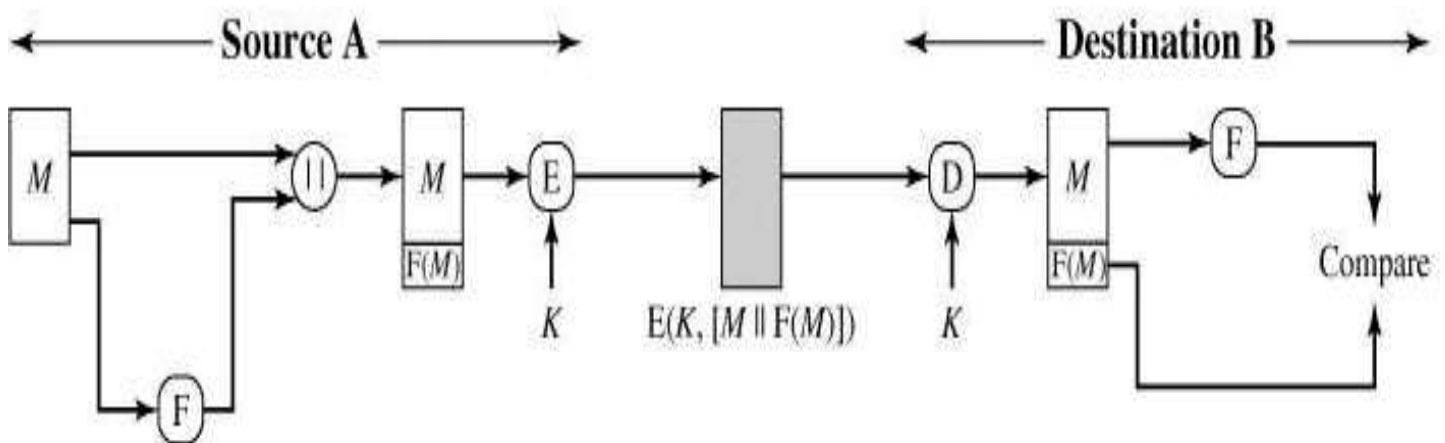
- Message encryption by itself can provide a measure of authentication.
- The analysis differs for symmetric and public-key encryption schemes.
 - Symmetric Encryption
 - Public-Key Encryption: Confidentiality
 - Public-Key Encryption: Authentication
 - Public-Key Encryption: Confidentiality and authentication

Symmetric Encryption



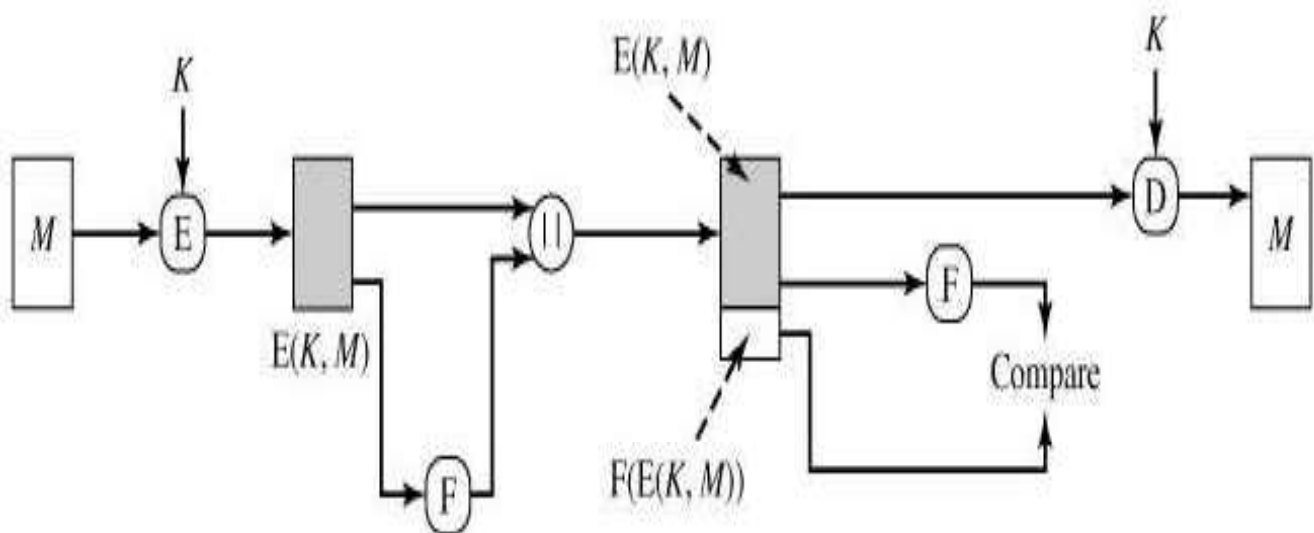
(a) Symmetric encryption: confidentiality and authentication

Internal Error Control - Append an error-detecting code, or frame check sequence (FCS) or checksum with the message



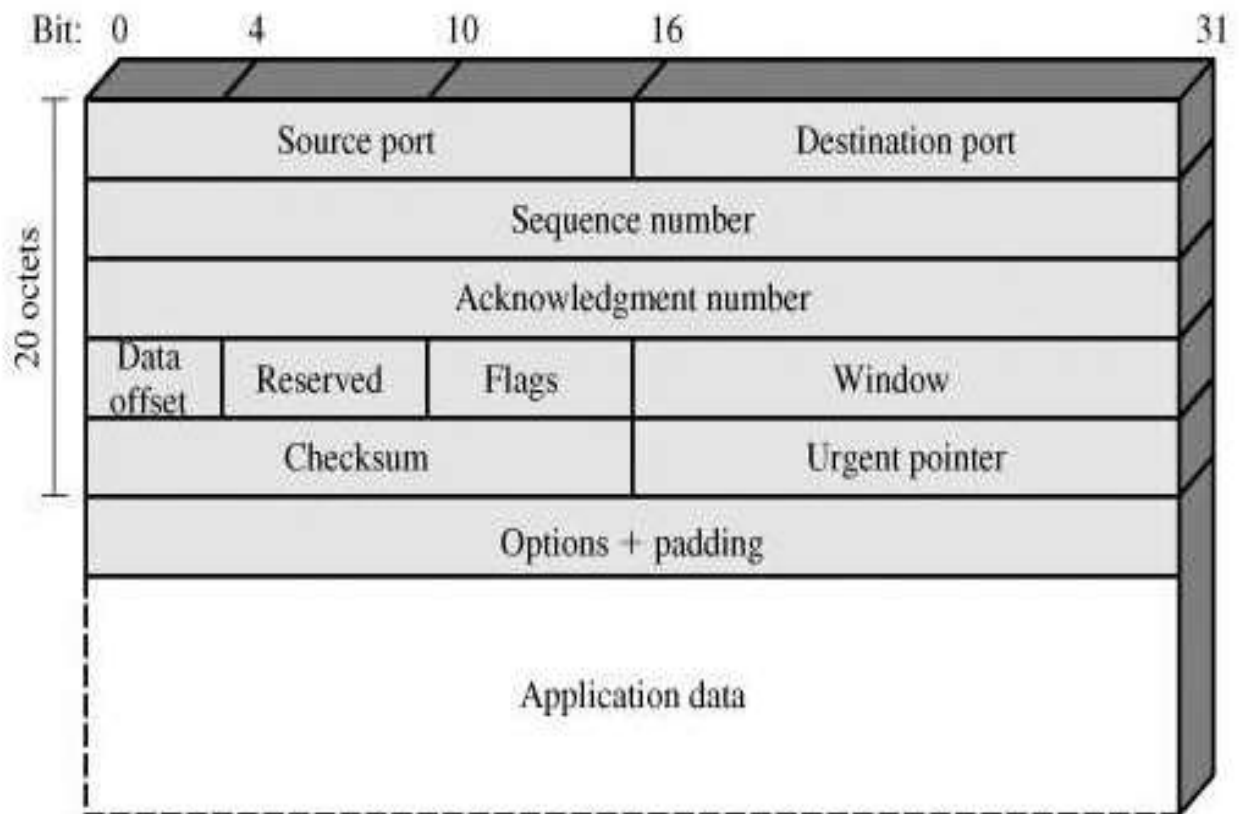
(a) Internal error control

External error control

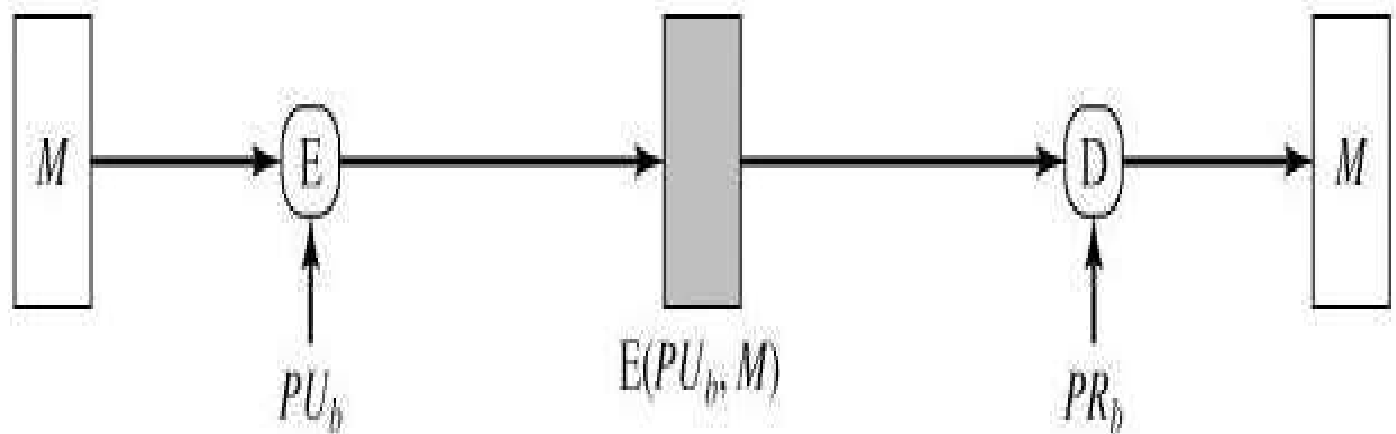


(b) External error control

TCP segment

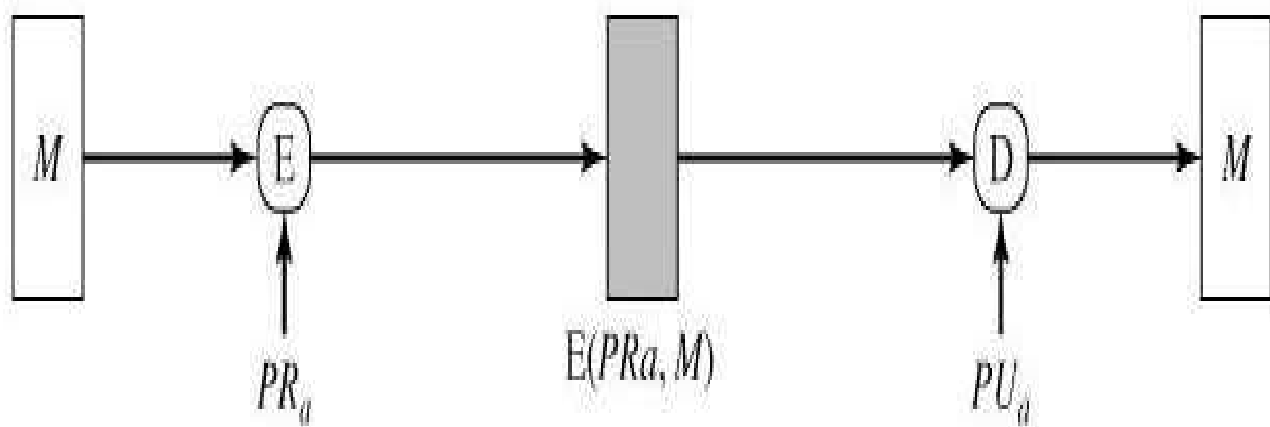


2. Public-Key Encryption: Confidentiality



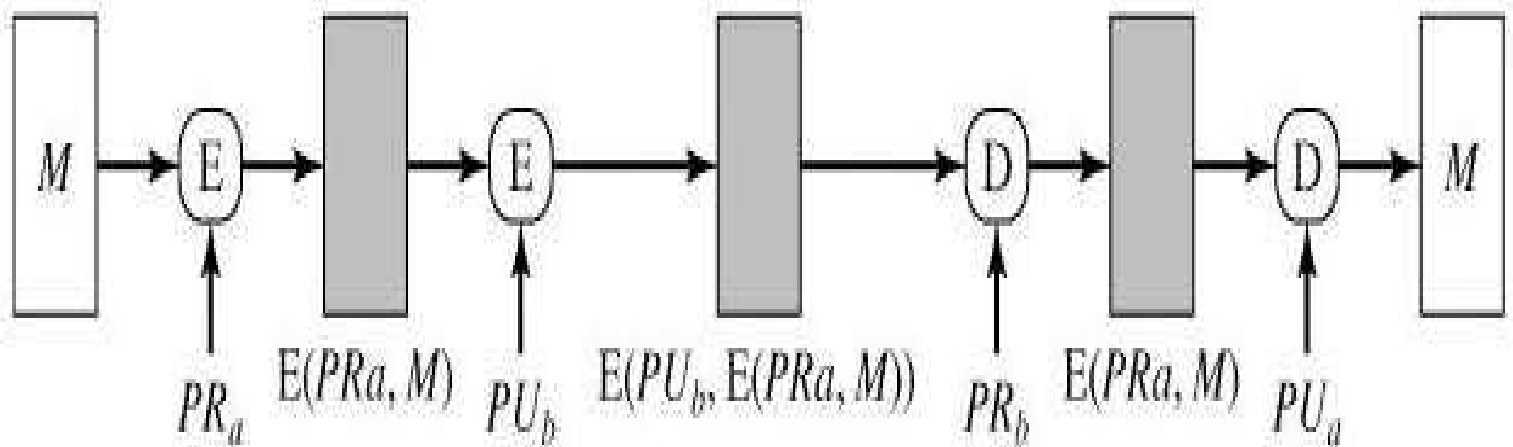
(b) Public-key encryption: confidentiality

3. Public-Key Encryption: Authentication



(c) Public-key encryption: authentication and signature

4. Public-Key Encryption: Confidentiality and authentication



(d) Public-key encryption: confidentiality, authentication, and signature

(ii) Message authentication code (1)

- An authentication technique involves the use of a secret key to generate a small fixed-size block of data, known as a **cryptographic checksum or MAC** that is appended to the message.
- two communicating parties, say A and B, share a common secret key K .

Message authentication code (2)

- When A has a message to send to B, it calculates the MAC as a function of the message and the key:

$$\text{MAC} = C(K, M)$$

Where $M = \text{input message}$

$C = \text{MAC function}$

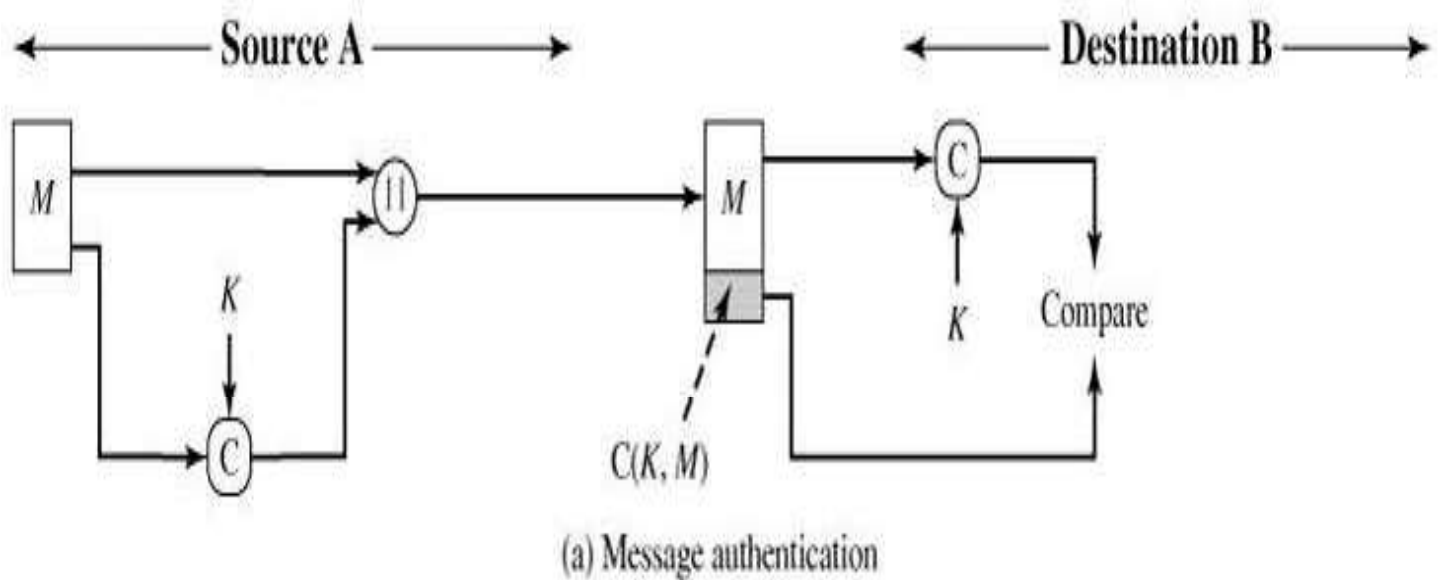
$K = \text{shared secret key}$

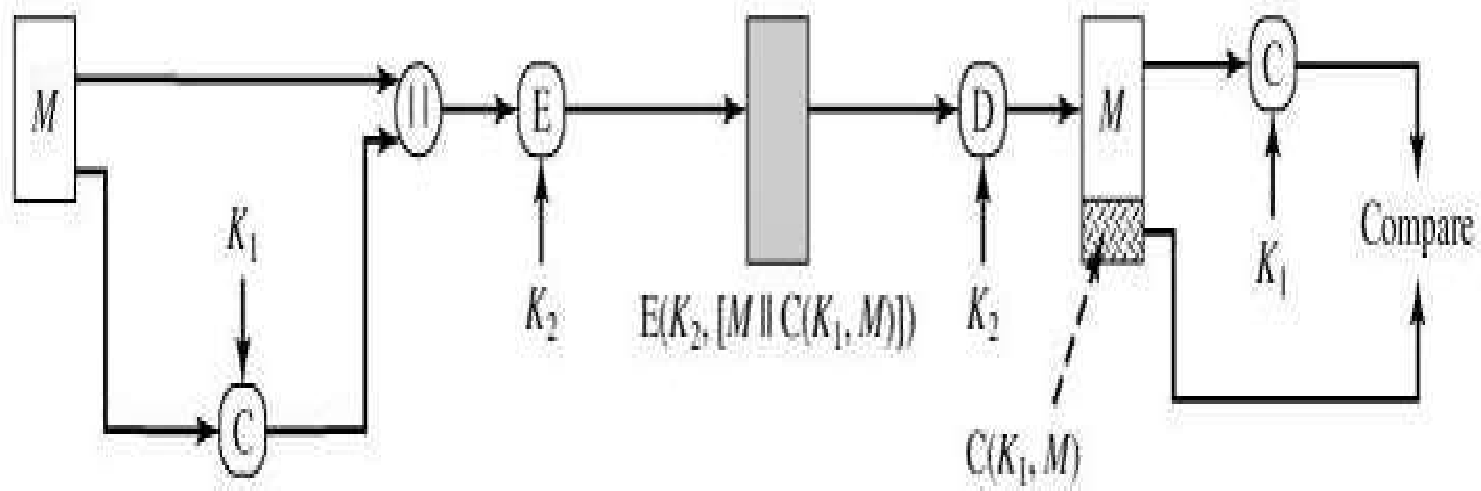
$\text{MAC} = \text{message authentication code}$

Message authentication code (3)

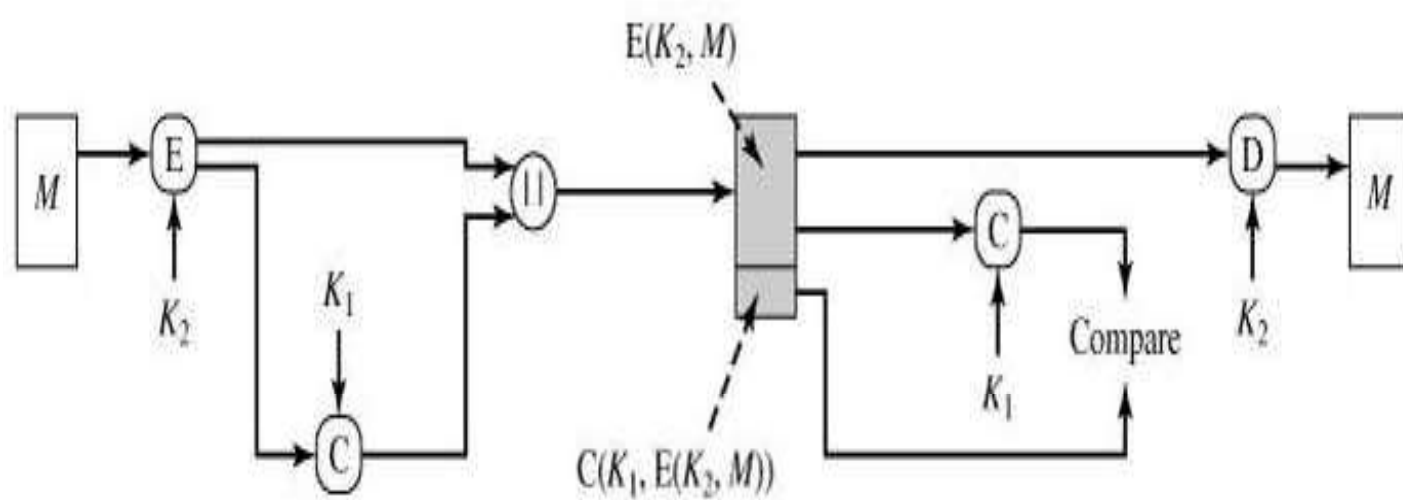
- message plus MAC are transmitted to the intended recipient.
- recipient performs the same calculation on the received message, using the same secret key, to generate a new MAC.
- received MAC is compared to the calculated MAC as shown in the figure

Message authentication code (4)





(b) Message authentication and confidentiality; authentication tied to plaintext



(c) Message authentication and confidentiality; authentication tied to ciphertext

(iii) Hash Function

- A variation on the message authentication code is the one-way hash function.
- As with the message authentication code, a hash function accepts a variable-size message M as input and produces a fixed-size output, referred to as a **hash code $H(M)$** .

- Unlike a MAC, a hash code does not use a key but is a function only of the input message.
- The hash code is also referred to as a **message digest or hash value**.
- The hash code is a function of all the bits of the message and provides an error-detection capability:
- A change to any bit or bits in the message results in a change to the hash code.

HASH FUNCTIONS

- A hash value h is generated by a function H of the form $h = H(M)$, where
 - M is a variable-length message
 - $H(M)$ is the fixed-length hash value.

- The hash value is appended to the message at the source at a time when the message is assumed or known to be correct.
- The receiver authenticates that message by re computing the hash value.
- Because the hash function itself is not considered to be secret, some means is required to protect the hash value.

Requirements for a Hash Function(1)

- The purpose of a hash function is to produce a "fingerprint" of a file, message, or other block of data.
- To be useful for message authentication, a hash function H must have the following properties:

Requirements for a Hash Function(2)

1. H can be applied to a block of data of any size.
2. H produces a fixed-length output.
3. $H(x)$ is relatively easy to compute for any given x , making both hardware and software implementations practical.
4. For any given value h , it is computationally infeasible to find x such that $H(x) = h$. This is sometimes referred to in the literature as the one-way property.

Requirements for a Hash Function(3)

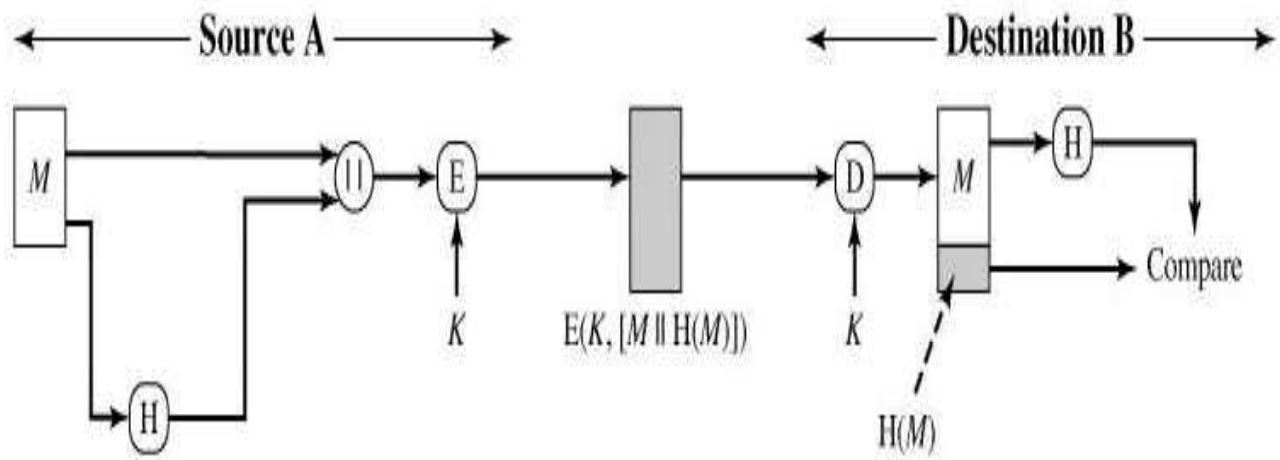
- For any given block x , it is computationally infeasible to find y not equal to x such that $H(y) = H(x)$. This is sometimes referred to as **weak collision resistance**.
- It is computationally infeasible to find any pair (x, y) such that $H(x) = H(y)$. This is sometimes referred to as **strong collision resistance**.

- The variety of ways in which a hash code can be used to provide message authentication, as follows:

a)

- The message plus concatenated hash code is encrypted using symmetric encryption.
- This is identical in structure to the internal error control strategy.
- The same line of reasoning applies: Because only A and B share the secret key, the message must have come from A and has not been altered.

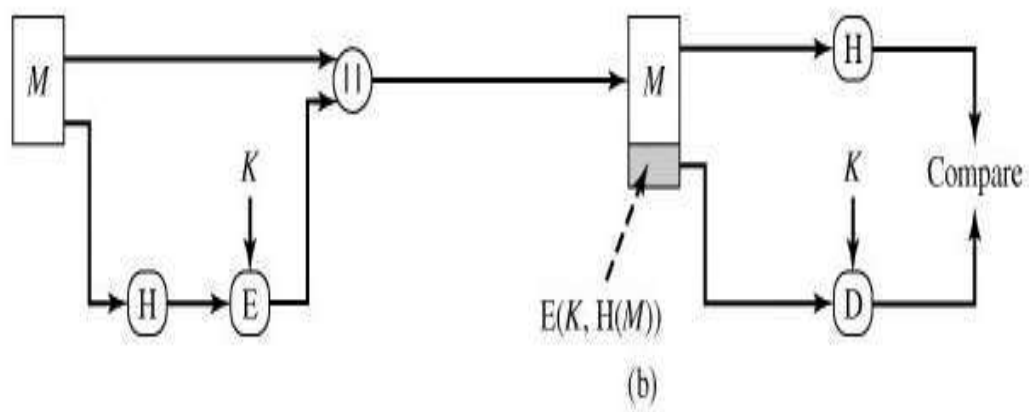
- The hash code provides the structure or redundancy required to achieve authentication.
- Because encryption is applied to the entire message plus hash code, confidentiality is also provided.



(a)

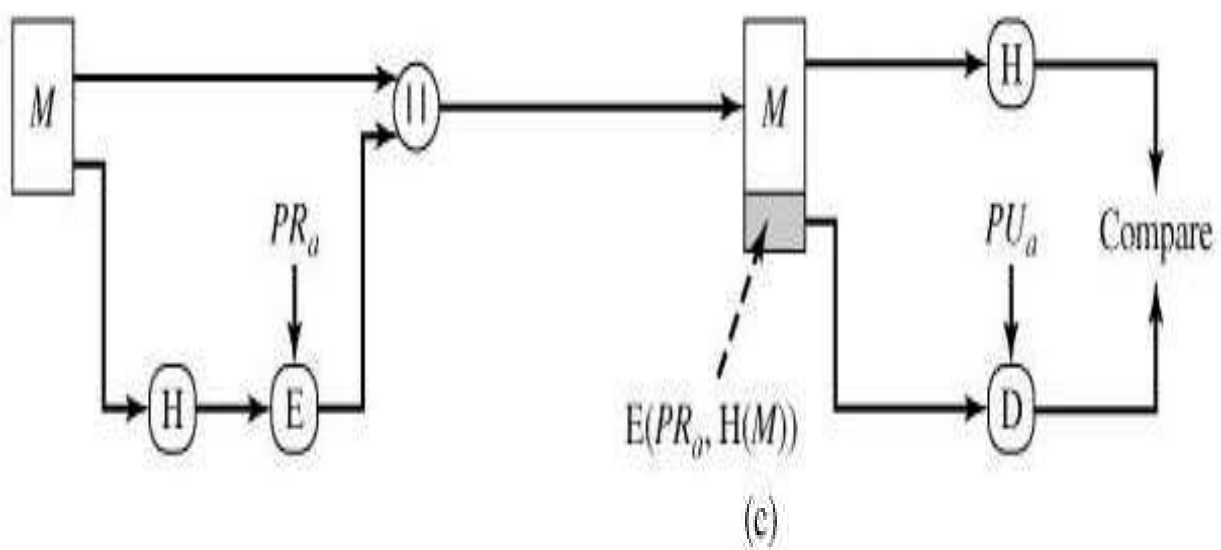
b)

- Only the hash code is encrypted, using symmetric encryption.
- This reduces the processing burden for those applications that do not require confidentiality.



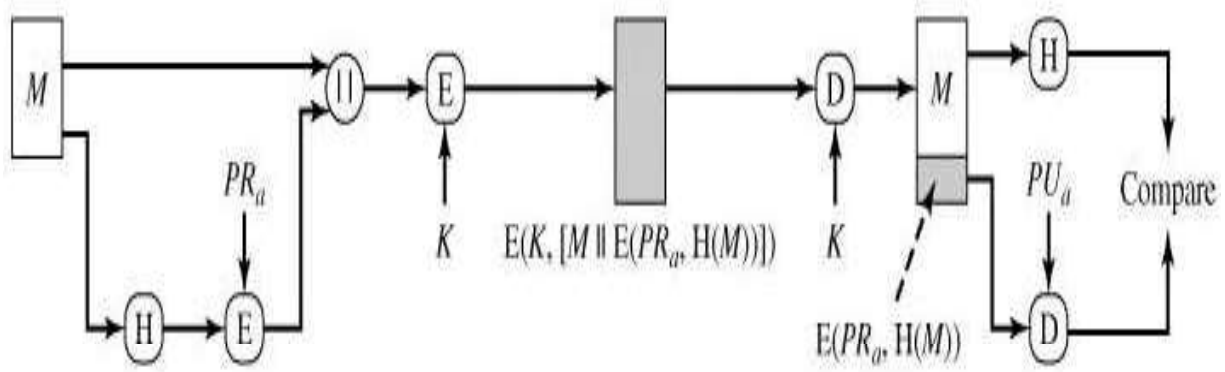
c)

- Only the hash code is encrypted, using public-key encryption and using the sender's private key. As with (b), this provides authentication.
- It also provides a digital signature, because only the sender could have produced the encrypted hash code. In fact, this is the essence of the digital signature technique.



d)

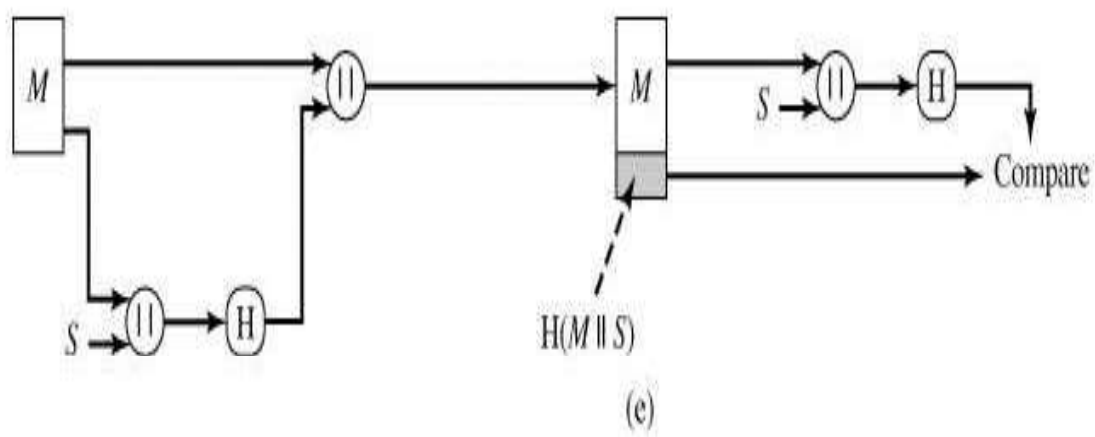
- If confidentiality as well as a digital signature is desired, then the message plus the private-key-encrypted hash code can be encrypted using a symmetric secret key. This is a common technique.



(d)

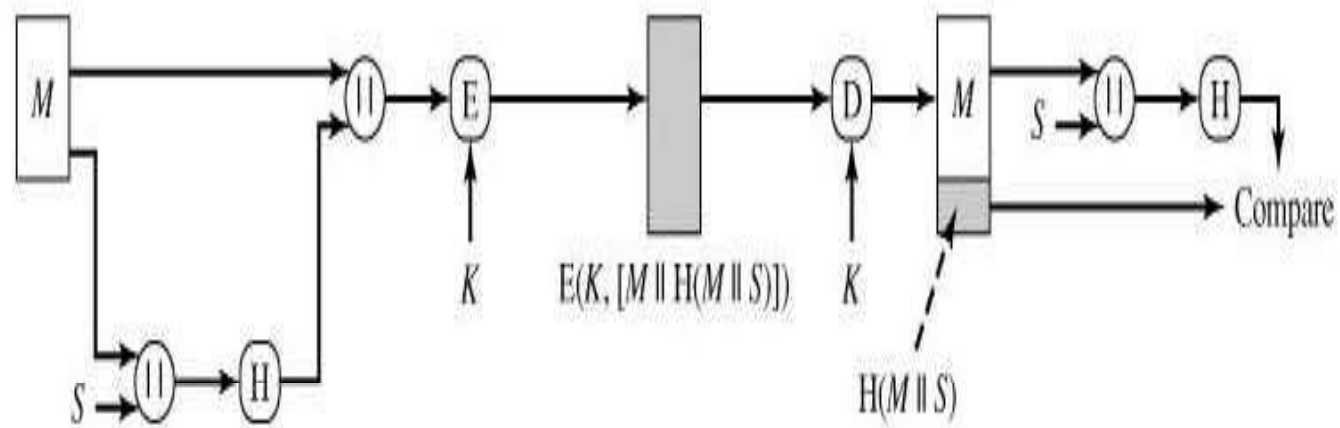
e)

- It is possible to use a hash function but no encryption for message authentication. The technique assumes that the two communicating parties share a common secret value S . A computes the hash value over the
- concatenation of M and S and appends the resulting hash value to M .
- Because B possesses S , it can re compute the hash value to verify. Because the secret value itself is not sent, an opponent cannot modify an intercepted message and cannot generate a false message.



f)

- Confidentiality can be added to the approach of (e) by encrypting the entire message plus the hash code.



(f)

SECURE HASH ALGORITHM

- developed by the National Institute of Standards and Technology (NIST).
- SHA-1 produces a hash value of 160 bits. With hash value lengths of 256, 384, and 512 bits, known as SHA-256, SHA-384, and SHA-512.

SHA-512 Logic

- The algorithm takes as input a message with a maximum length of less than 2¹²⁸ bits
- and produces as output a 512-bit message digest.
- input is processed in 1024-bit blocks.